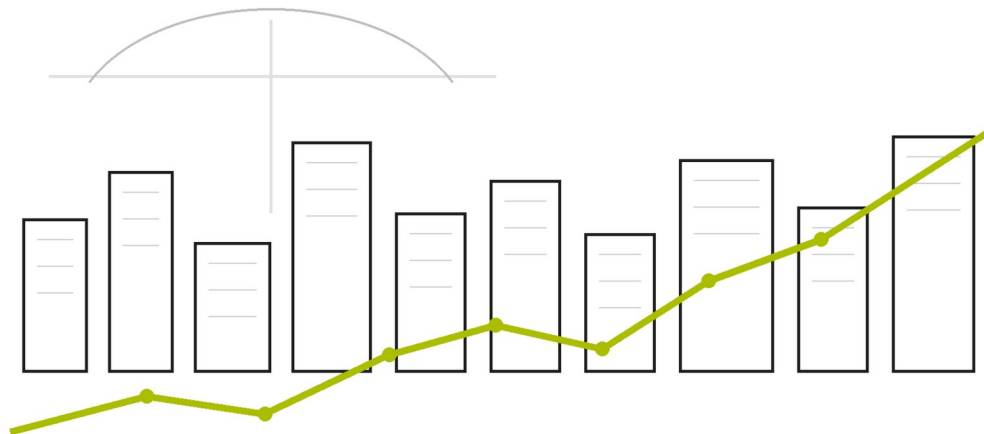




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From Brasília to the Plant Floor

People, Data and AI in Sustainable Digital Transformation



*"Everyone wants the shiny city on the hill.
The road to get there is paved with data."*

EDUARDO SOPALDA

Global Data Governance Lead · dsm-firmenich

01. THE ARCHITECT

I am an architect. Not as a metaphor. I studied architecture. I drew buildings. Then I spent the last twenty years designing something else: data and AI governance for large organisations. Today I sit in Digital and Technology at dsm-firmenich, the team that runs every data and AI initiative in the company, and I lead governance across all of our business units, sustainability included. So there is a slightly strange continuity in my career. I moved from designing physical spaces to designing the ethereal structures around data, but I keep seeing the same problem. We design from above, not for the person who has to live with it.

That matters when we talk about sustainable digital transformation, because sustainability is full of ambitious plans, targets, systems and increasingly sophisticated technology. But every one of those things eventually has to meet a human being. Someone has to extract the data, interpret it, challenge it, trust it and sometimes tell the system that it is wrong. I am not here to sell you a framework this morning. I work inside a chemical company, like many of you, and I have got plenty of these things wrong myself. What I want to explore is the distance between the beautiful plan and what actually happens at ground level.

02. BRASÍLIA

Let me take you to Brasília. It is one of the most extraordinary examples of designing from above. Lúcio Costa created the urban plan and Oscar Niemeyer designed many of its most famous buildings. From the air it is monumental, precise and unforgettable. Jan Gehl later coined a name for what he saw there: the Brasília Syndrome. Plans drawn by people looking down from above, then dropped onto the ground with little regard for the people who have to live there. A city can make perfect sense from a helicopter and feel completely different when you walk through it.

The distances are enormous, the spaces are difficult to navigate on foot and there are very few of the small human spaces that make a city feel alive with human encounters. I am not saying Brasília is a bad city. That is not the point. The point is scale. A plan can be beautiful and still fail to account for the experience of the person standing inside it.

And there is one detail I want you to sit with. Many of the people who built Brasília could not afford to live in the city they had built. They ended up commuting in from satellite towns. The city was finished. It just was not designed around the people who had to live and work in it.

03. WHEN THE DRAWING MEETS THE GROUND

We build governance in a very similar way. We have policies, controls, committees, operating models and, occasionally, a beautiful RACI that makes everyone feel that civilisation has finally arrived. Seen from above, from the auditor's chair, the regulator's seat or the steering committee, everything can look symmetrical and complete.

Then the framework reaches the plant.

The policy may be too long to cross. The process may be too wide to walk through on a Tuesday morning when the plant is running. And there may be no space where a process engineer and a data owner discover that they have been using the same word to mean two different things for three years.

I have been the person with the beautiful drawing. I have presented the framework, watched people nod, and then watched them carry on exactly as before. They were not being difficult. I had built them a plaza nobody wanted to stand in.

So I stopped being interested only in whether a framework was complete. I became much more interested in whether people actually used it.

04. SUSTAINABILITY HAS THE SAME PROBLEM

This is where sustainability enters the story for me. We are very good at designing sustainability from above. We set ambitions, targets and roadmaps. We talk about emissions, responsible sourcing, circularity, product footprints and increasingly about digital tools that can help us monitor and automate all of this.

SPEAKING SCRIPT

But eventually the ambition has to become evidence.

At dsm-firmenich, sustainability is expressed through People, Planet and Progress. That is important to me because it puts the human element inside the sustainability story rather than somewhere outside it. The ambition may be global, but the data that makes that ambition measurable is created across businesses, processes, systems, suppliers and people.

That creates a very practical question. How do you make sure that the information travelling through all of those places still means the same thing when it reaches the other end?

Because sustainability reporting is not only about producing a number. It is about everything standing underneath it. Where did it come from? Who owns it? What definition sits behind it? What happens when two systems disagree? And who is prepared to stand behind it when somebody asks a difficult question?

That is where sustainability and governance stop being two separate conversations.

05. THE CARBON NUMBER

A carbon number looks like chemistry. It is not. It is a data product.

Behind one apparently simple number can sit supplier declarations, energy meters, batch records, transport data, allocation rules and a collection of assumptions. Much of that information comes from systems that were never designed to be added together, and suppliers may use their own definitions for the same words.

So the challenge is not simply calculating the number. The challenge is being able to explain that number.

Once that information becomes part of sustainability reporting and assurance, lineage, ownership and shared definitions stop being data hygiene. They become part of the evidence.

And this is where the human part becomes very real. Someone has to know which source is right. Someone has to understand why the number changed. Someone has to challenge an assumption. Someone has to be able to explain the result to another person who was not in the room when the original decision was made.

You cannot report what you cannot trace.

06. THEN SOMEONE SAYS: WHY DON'T WE JUST USE AI?

And then, inevitably, someone says: why don't we just use AI?

I am optimistic about AI, so I am not going to give you the usual speech about why we should all be frightened of it. But I do think we sometimes ask AI to solve problems that existed long before AI arrived.

If five people in an organisation use the word batch to mean eight different things, putting an AI agent on top of that ambiguity does not remove the problem. It can simply scale it up.

The same applies to the data we give these systems. In one agent project I saw, around ten thousand documents went in and roughly two thousand were actually relevant. The rest were duplicates, superseded drafts or the same deck sitting under four different filenames. All of it was embedded. All of it was consuming tokens and energy.

So filtering before you embed is not merely an AI optimisation. It is a governance decision, a cost decision and a sustainability decision at the same time.

Sometimes AI is the answer. Sometimes the answer is better data. Sometimes it is a better process. And occasionally the answer is simply agreeing on what a word means before we ask a machine to interpret it.

07. THE ROAD TO THE SHINY CITY

Everyone wants the shiny city on the hill. We want the autonomous plant, the perfect digital twin, real time sustainability intelligence and AI agents that help us make informed decisions at speed. It is an exciting destination.

But we do not get there by teleportation.

SPEAKING SCRIPT

The road to that city is paved with data. And not the glamorous kind of data. It is paved with definitions, ownership, lineage, quality and context. It is paved with people who know what a number actually means and are willing to use the systems that depend on it.

This is the part of digital transformation that is easy to underestimate. We talk about the destination because it is exciting. We talk about the technology because it is visible. But the real work is often underneath, in the data foundations and the human behaviours that make the technology usable.

If that road is built from data nobody trusts, definitions nobody shares and processes nobody adopts, the city on the hill does not matter. We have simply built another Brasília: impressive from above, but disconnected from the people who have to operate it.

08. BACK AT GROUND LEVEL

Take digital twins. Everyone in this room knows roughly what one is: a digital representation of a physical asset or process, fed by equipment data, sensors, production systems, laboratory results, maintenance records and energy consumption.

The interesting question is not what a digital twin is. It is whether anyone believes it.

AI can help build, enrich and operate these models. It can find patterns in large volumes of operational data, predict behaviour and support the calls that people have to make. But using AI does not automatically make the twin sustainable.

The sustainability comes from what the system lets us understand and improve. If the twin shows how a process really behaves, it shows where energy is being consumed, where material is being lost, where emissions are being generated, or where the process could be run differently. Sustainability becomes part of the way we operate rather than something we measure only afterwards.

And that brings us back to governance. The twin is only as good as the information flowing into it. If the equipment data is incomplete, if the definition of yield changes between systems, if energy data is attributed incorrectly, or if nobody trusts the model, the representation on the screen is still only a representation.

It becomes sustainable, in both senses, when the operator on the night shift trusts what it tells them, when the engineer understands which definition of yield sits behind the model, and when somebody is accountable if the system is wrong.

That is also why I think governance has to be designed at human scale. A framework can be elegant. A digital twin can be technically impressive. An AI system can be extraordinarily capable. But if the person using it does not understand it, trust it or know what to do when it is wrong, then we have not really transformed the process. We have added technology to it.

Which brings me to the principle I keep coming back to as these systems become more autonomous.

Agents do not own risk. They do not carry accountability. Humans do. Always.

09. BACK TO BRASÍLIA

So let me take you back to Brasília.

The city is still there. Still beautiful from the air. Still very different at street level. It is an analogy, nothing more. The real question is whether we can build the sustainable digital transformation we are describing without losing the people who have to make it work.

I think we can. And I think it is worth doing.

We are all well aware of the stigma around AI: its energy consumption, its infrastructure requirements and the environmental questions that come with scaling these systems. Those questions are real. But there are also people deliberately building these tools to help slow global warming, using AI not only as something that consumes resources, but as something that helps us understand and address climate challenges.

That is how I think we should approach this. Not by pretending the technology has no cost, and not by rejecting it because it does. By understanding what it takes to build it responsibly, where it can create genuine value, and how governance can help us make those choices deliberately.

SPEAKING SCRIPT

Frameworks are necessary. They are just not sufficient. The opportunity is to build systems that are useful, trusted and sustainable in practice.

10. THE QUESTION

So I want to leave you with one question.

What is one word in your organisation that means something different depending on which floor you are standing on?

Ask it at lunch.

You may discover that the answer tells you more about your governance than another hundred pages of framework ever could.

Thank you.